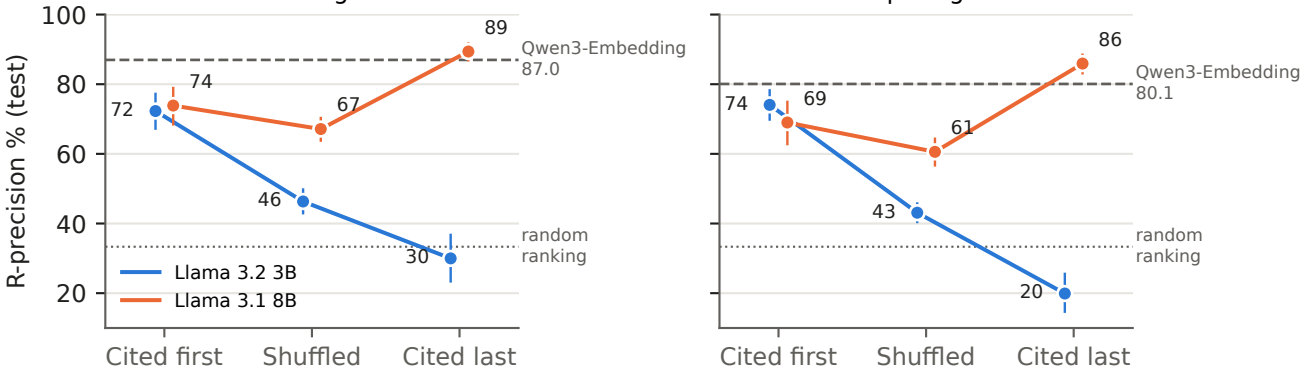
random negatives
concept negatives
R-precision % (test)
100
80
60
40
20
Qwen3-Embedding
87.0
Qwen3-Embedding
80.1
random
ranking
random
ranking
Llama 3.2 3B
Llama 3.1 8B
72
74
67
89
46
30
74
69
61
86
43
20
Cited first
Shuffled
Cited last
Cited first
Shuffled
Cited last